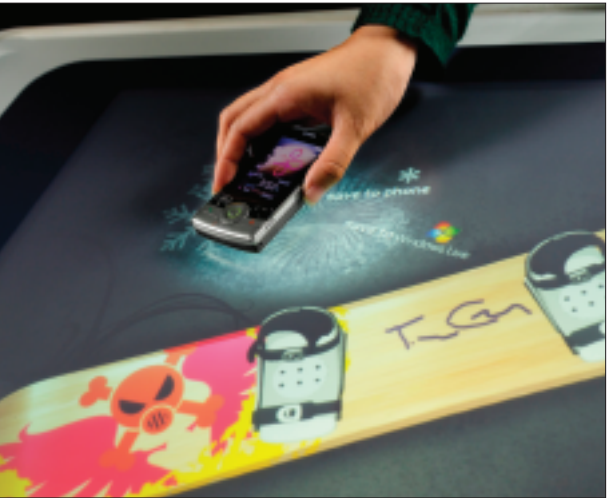


(TouchWall) and the Vista-based software called Plex. TouchWall has some superficial similarities with Surface, in that both systems permit multi-touch interaction with the interface. What differentiates the TouchWall from the Surface, though, is that it is a much simpler solution. It consists of three infrared light sources, which scan the entire surface. The surface in this case, consists of a 4x6 foot plexiglass sheet. When something hinders the infrared beams, a camera notes the position and feeds the information to the software heart—Plex.



Surface features a 30-inch tabletop display which allows several people to work independently or simultaneously, without using a mouse or a keyboard

TouchWall was recently demonstrated by Microsoft Office Labs General Manager Chris Pratley and Director of Envisioning (nice title) Ian Sands. Prately envisioned that hopefully in the near future, computing would be less monitor / keyboard / mouse and more like an architect's desk, with direct interactions via voice commands and touch surfaces.

Sands demonstrated a number of activities to that effect—interacting with media with his hands, zooming in and out of Web pages, and using drawing tools to turn the entire screen into a whiteboard.

The relatively simple hardware used for TouchWall—one camera, three infrared laser emitters and a plexiglass surface should make the product more pocket-friendly than the Surface initiative. Microsoft suggested a price in the range of a 'few hundred dollars'.

More exciting than the multi-touch, tactile interactions offered by these products is the user interface evolution that this change in hardware is destined to trigger. The iPhone, for example, has re-imagined the way we interact with a phone or even a PDA. Gone are ugly menus so common with Windows Mobile devices, or pad-driven interfaces which are the hallmark of almost every cell phone. The iPhone brought

along a rich, high-resolution, icon-driven interface. Viewing photographs no longer meant a long list of thumbnails to scroll through, but full-screen images that you can flip through, much like you would a photo-album. On the other hand, the Microsoft Surface has pioneered multiple device interactions. Downloading videos and photographs from a camera to a personal computer is done by simply placing the camera on the Surface. One probable use of the Surface in the hotel industry explores the possibility of placing a tagged wineglass on the surface, only to trigger the Surface to fan out all the wine offerings of the hotel, around the bottom of the glass. Another possible use explored by T-Mobile, is to display specifications of a cell phone around it when placed on the Surface—making comparisons between cell phone models a breeze. The TouchWall, too, brings along a unique interface—resembling an infinite work table. Electronic documents can be stacked, much like your office desk. Browsing through your presentations is then as simple as touching this stack to zoom in and then making page flipping gestures to move between pages. The central thrust of the TouchWall concept is to organise and view content spatially. Zoom out to a stack of your videos, touch to zoom in, and tap to play the video, zoom out again to a Web page, zoom in to check your mail, zoom out to a document you wish to mail, and so on. The status quo of trudging through file managers, sub-sub-sub folders, menu structures and icons is done away with, through such an interface.

Of Screaming At Windows And Winking At Hyperlinks

What else can we expect down the years, as either an evolution or a revolution of the man-machine interface? The difficulty lies in the prediction. The usual suspects have been circling the aging mouse-keyboard duo for ages now, to no avail.

Voice-Activated Interfaces: The staple of science fiction—talking directly to a machine to carry out your whims and fancies. Not science fiction anymore though. Dictation software that jot down spoken words to your favourite word processor have been around for quite sometime. Current operating systems such as Windows Vista even come with rudimentary voice command interfaces. But these are baby-steps. Software is not so smart that we can carry a natural conversation with a machine such that it understands our vocal idiosyncrasies to even take down notes, let alone carry our complicated tasks. Then there is the hurdle of language itself—there are so many ways we humans speak. One day (as Bill Gates loves to remind us almost every year), voice-recognition and voice-command will indeed become

a viable means of communicating with our computing machines.

Brain-Machine Interfaces: This idea is still fantastical, although it has already hatched and taken its first primordial steps in laboratories the world over—the human brain interfaced directly to machines. Perhaps the ultimate interface—think and the mail is typed and sent, blink and you are at your favourite online shopping portal, sneeze and your computer gets a virus... maybe not the last bit, or so we hope. Brains have been tested to drive machines for decades now—from rats to humans. The biggest human success stories and research are done to ease paraplegics who are 'locked' in their bodies. These stories range from research to ease the disabled move a wheelchair with just their thoughts, or to communicate via a computer by blinking, and so on. Such mind-machine links are generally done in one of two ways—electrodes are either linked to the surface of the cortex, in a direct machine-brain connection, or are placed on the surface of the head, and run on the same brain firings that an EEG machine detects and analyses. Both these methods have met limited success. People have been able to do little more than move boxes around a computer screen, or a mouse cursor. A more practical implementation of an EEG-running interface was released in the form of a baseball cap. A team of researchers from Taiwan designed a brain monitoring system inside a baseball cap. The wireless and portable system, can process data and provide feedback in real time. It is currently being tested as an early-warning system to warn drivers that they are falling asleep on the wheel. In future, similar technology could very well be used to allow you to control home appliances such as TVs, computers, and air conditioners, all by just thinking about them! As you can imagine though, such interfaces are still a long way from bearing fruit.



A tetraplegic patient attempts left or right hand movements and tries to move the circle from the middle of the screen to the target
Copyright © 2005 by LCE.

Eye-Tracking Software: A few months back, a small video on YouTube created a lot of buzz in the tech circles. Created by Johnny Chung Lee, from Carnegie Mellon, it showed him using the Nintendo Wii's controller to navigate around a 3D space. What was so special about that? The navigation was done by Chung Lee simply moving his head around, while the IR sensor in the Wii Remote tracked him. The tech behind the seeming magic is based on IR. Lee wore glasses with small IR transmitters tacked on the sides. The Wii Remote controller tracked these IR points through its IR window and some clever software did the rest.

A similar experiment was performed by a Stanford researcher, Manu Kumar. In March 2007, Kumar successfully demonstrated an alternative to the mouse. His eye-tracking interface allows you to remote control a computer interface using movements of the eye. The interface allowed the person to click hyperlinks, highlight text, and scroll simply by looking at the screen and tapping a key on the keyboard. Manu Kumar's eye-tracking software had some common elements with Johnny Lee's—a high-definition camera to do the actual tracking and infrared light served as the tracking points.

Eye-tracking and head-tracking software seems feasible, especially given the fact that monitors today come armed with cameras embedded anyway. However, the resolution for cameras would have to be much higher to successfully track a reference point. Perhaps some day soon, we will be able to close a window just by staring at it in anger...

Where Do We Go From Here?

Multi-touch has made decent inroads into our interactions with computing devices. Considering that the technology went public only a year ago, it is a good show. But what else can we look forward to? Will the Microsoft Surface make it to our living room anytime soon? Or will the TouchWall make an appearance in our office meeting rooms? Would our laptops allow us to interact with our operating systems by just blinking at them, or by moving our heads? Only one thing is certain, and that's uncertainty. Predictions prove futile in this fast-moving tech world of ours. It will only take one successful product, launched by a visionary company to completely change the way we interact with data. Whether this would-be product is based on touch sensitive surfaces, brain-machine interfaces, or on voice-commands, no one can tell for sure. It might as well run on technology not accounted for in this article... What is certain is that we could do a lot better than what we have today. The days of the tacky keyboard and mouse, and the menu and icon based user-interface that they have spawned, must surely be numbered. ■

readersletters@thinkdigit.com